(a shorter time between pulses) than the frequency with which they were emitted.

Box 2.4

Gravitational Time Dilation

Take two identical clocks. Place one on the floor of a room beside a hole into which it later will fall, and attach the other to the room's ceiling by a string. The ticking of the floor clock is regulated by the flow of time near the floor, and the ticking of the ceiling clock is regulated by the flow of time near the ceiling.

Let the ceiling clock emit a very short pulse of light whenever it ticks, and direct the pulses downward, toward the floor clock. Immediately before the ceiling clock emits its first pulse, cut the string that holds it, so it is falling freely. If the time between ticks is very short, then at the moment it next ticks and emits its second pulse, the clock will have fallen only imperceptibly and will still be very nearly at rest with respect to the ceiling (diagram a). This in turn means that the clock is still feeling the same flow of time as does the ceiling itself; that is, the interval between its pulse emissions is governed by the ceiling's time flow.

